

IIT Bombay – Honeywell Centre of Excellence for Future Skills & Innovation

BenefitBuddy AI

“One Profile Every Benefit You deserve”

An AI-powered platform that helps every citizen discover, compare and apply for the government schemes and insurance plans they are eligible for.

Team Name: Superstars

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1. Summary

Millions of citizens in India don't receive the government welfare benefits , subsidies and insurance protection they are entitled to not because they are ineligible but because they don't know these

schemes exists or don't know they qualify . BenefitBuddy AI is a single digital platform that solves both halves of this problem : it tells a user , based on a short profile (age , category , income , landholding , occupation and more),exactly which government schemes they are eligible for and what they stand to receive and also lets them compare every available health and life insurance option side by side so they can pick the plan that genuinely fits their needs and budget.

Unlike PolicyBazaar and myScheme.gov.in our application provides a single personalized profile for discovering , assessing eligibility .This addresses critical information asymmetry for underserved populations , promoting financial inclusion and poverty alleviation in line with SDGs.

2. Problem Statement

- i. Awareness & Fragmentation : many of the eligible citizens remain unaware of government benefits cause information is scattered across dozens of decentralized ,opaque portals with inconsistent eligibility logic.
- ii. Insurance overload : When people do try to protect themselves through health or life insurance they face an overwhelming number of products from different insurers with the inconsistent terms , making it hard to identify the best value option.

Result : Underutilized welfare budgets, low insurance penetration and continued vulnerability among exactly the population these schemes and products are meant to protect.

2.1 Why Citizens Miss Benefits Despite These Portals Existing

There are reasons why people miss out on benefits.

- These portals are not user-friendly. A person has to know what they are looking for.
- The eligibility rules are written in a way that's hard to understand.
- People often find out about these schemes through word of mouth, which is slow and unreliable.
- no platform shows government schemes and private insurance side by side.
- Many people do not speak English. They are not comfortable using digital platforms.
- A person has to upload the documents for every separate application.

3. **Proposed Solution**

Our project platform is built around 2 integrated modules

- Government Scheme Eligibility Engine - user enters these basic details and instantly receives a personalized ranked list of schemes they qualify for .
- Insurance Comparison Engine – the user browse and compares health and life insurance products across providers on cover amount , premium , tenure and claim – settlement track record , so they can choose the option that best matches their needs and budget.

3.1 How the AI Adds Value Beyond a Static Portal

- BenefitBuddy AI matches schemes to a person by listing schemes.
- The eligibility rules are explained in language.
- The recommendations update as the profile changes.
- The chatbot can answer questions in language.
- Government insurance and private insurance plans are compared side by side.

3.2 User Journey. A Farmer

- Let's take the example of a farmer named Ramesh. He owns 1.5 acres of land in Maharashtra. Grows cotton. He signs up on his phone. Fills in his profile.
- The engine flags him as a farmer and matches him to the PM-KISAN scheme.
- It also shows him the Pradhan Mantri Fasal Bima Yojana for crop insurance.
- The insurance comparison screen shows him two or three crop/health add-on plans.
- The document checklist tells him what documents he needs to apply.
- His Benefit Score shows him what schemes he is currently using and what he is eligible for.

3.3 User Journey. A Student

- Let's take the example of a student named Priya. She is 19 years old and studies in Odisha. She enters her education level, category, family income, and state.
- The engine matches her to the National Scholarship Portal schemes.
- It flags a central-sector scholarship scheme based on her marks and income slab.
- Smart Notifications alert her when the application window opens.
- Her AI Action Plan tells her to apply to the matric scholarship first

4. Key Features

- AI eligibility engine for schemes based on demographics , land etc
- Comprehensive Insurance Comparator
- Personalized Dashboard
- Application Tracking
- Notification Engine

5. Competitive Landscape and Differentiation

Capability	Bennefit Buddy AI	Policy Bazar – type platforms	myScheme portal
Government Scheme eligibility	✓	×	✓
Health /life insurance comparison	✓	✓	×
One profile covering both schemes and insurance	✓	×	×
End to end application support	Guided in app assistance planned	Not applicable to schemes	Redirects users out to separate schemes/ministry portals to actually apply
Rural / low-literacy accessibility	Voice interaction and regional language UI planned	Largely on urban centiric English/Hindi and app UX	Standard government web
Renewal / new-scheme alerts across both categories	Unified remainder	Renewal remainders	NO proactive alerting for new matching schemes

6. How it Works

- Step 1: User creates a profile and enters demographic and financial details.
- Step 2: The Eligibility Engine matches the profile against the scheme database and shows a ranked, easy-to-read list of qualifying schemes with expected benefit amounts.

- Step 3: If the user is interested in insurance, they open the Comparison module and filter by cover amount, tenure, and budget.
- Step 4: The platform presents a clear side-by-side comparison table of matching insurance plans.
- Step 5: The user saves shortlisted schemes/plans to a personal dashboard and receives reminders for deadlines, renewals, or newly launched schemes matching their profile.

7. Prototype

7.1 Profile Intake Screen

BENEFITBUDDY



Welcome back

Tell us a bit about yourself to discover government benefits you may qualify for.

Full Name
Sameera

Age
20

Gender
Female

Annual Family Income (₹)
600000

State
Delhi

Category
General

Occupation
Student

Land Holding (Acres) - Optional
e.g., 1.5 (Leave blank if none)

Analyzing Eligibility...

A single form captures everything the Eligibility Engine needs — age, gender, category, occupation, annual family income, state, and optional land holding — replacing the dozens of separate forms citizens currently fill out across scattered portals. The moment a user submits, the profile is scored against every scheme in the database simultaneously.

BENEFITBUDDY



Welcome back

Tell us a bit about yourself to discover government benefits you may qualify for.

Full Name
Ravi

Age
60

Gender
Other

Annual Family Income (₹)
25000

State
Punjab

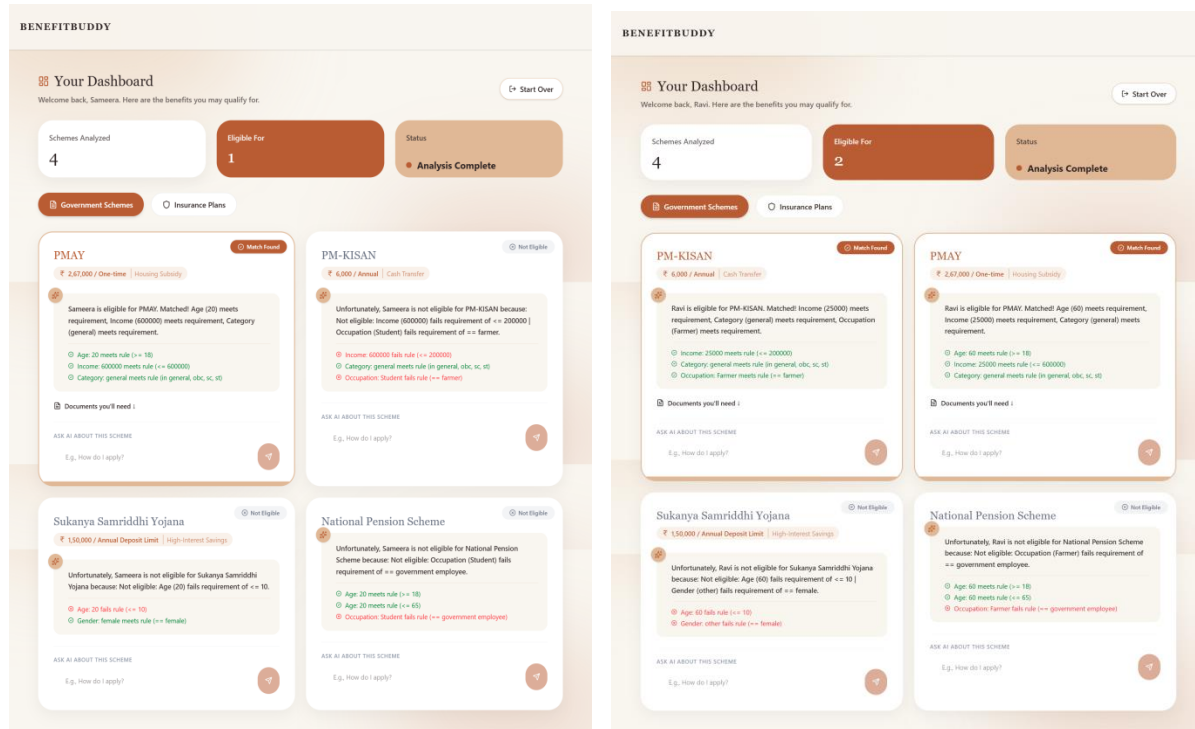
Category
General

Occupation
Farmer

Land Holding (Acres) - Optional
1.8

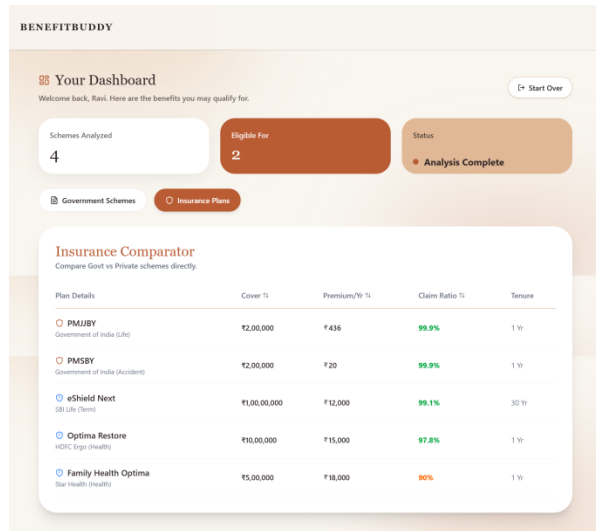
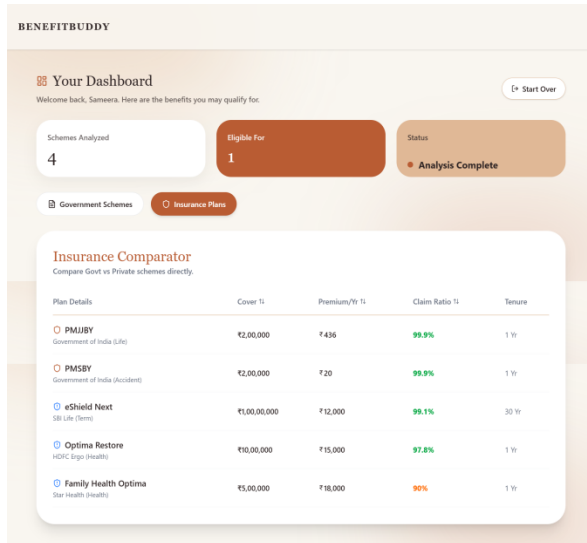
Analyzing Eligibility...

7.2 Scheme Eligibility Dashboard



Rather than a plain "eligible/not eligible" flag, every match shows its reasoning — each criterion (age, income, category, occupation) is checked individually and displayed with the exact rule applied. A rejected scheme like PM-KISAN explains precisely why (income and occupation both fail the threshold), while a match like PMAY shows every passing condition along with the expected benefit amount. An "Ask AI" box on each card lets the user query that specific scheme in plain language.

7.3 Insurance Comparator

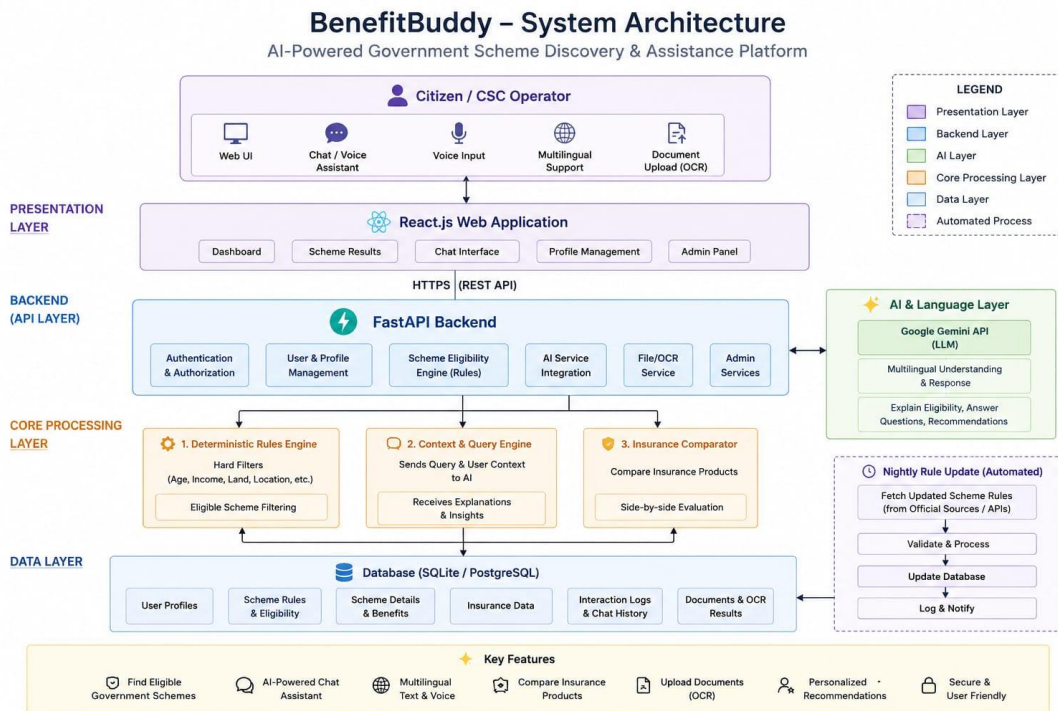


The same profile feeds directly into a side-by-side comparison of government and private insurance plans — cover amount, annual premium, claim-settlement ratio, and tenure — sorted so the strongest value is easy to spot. Government schemes (PMJJBY, PMSBY) sit alongside private options (SBI Life, HDFC Ergo, Star Health), giving the user one screen to evaluate both instead of visiting separate insurer sites.

8. Technical Approach

Our build uses tools proven out in our team's prior projects, adapted to the eligibility-matching and insurance-comparison problem:

8.1 System Architecture



- **Backend/Frontend:** FastAPI backend exposing REST endpoints for profile intake, eligibility scoring, and insurance comparison; React frontend for the user-facing dashboard.
- **Data layer:** A structured scheme/insurance database (state + central schemes, insurer products) normalized into a common eligibility-rule schema, refreshed from official portals (myScheme.gov.in, state welfare portals, IRDAI-listed insurer filings).
- **Eligibility Engine:** A hybrid engine — a deterministic rules layer encodes hard eligibility criteria (age, income slab, land ownership, category) for auditability, while an XGBoost ranking model learns from user outcome data to rank matched schemes by likely relevance and benefit value, rather than showing an unordered list.

- **AI/NLP layer:** Gemini API powers the regional-language chatbot layer — explaining eligibility rules in plain language, answering follow-up questions, and voice-interaction support for low-literacy users.
- **Insurance Comparator:** A rules-based comparator scores insurance products on cover amount, premium, tenure, and claim-settlement ratio, surfaced as a side-by-side table.

8.2 Feasibility Within Hackathon Scope

- **Build in hackathon window:** working profile intake, rules-based eligibility engine for a limited set of schemes (e.g. PM-KISAN, National Scholarship Portal, PMFBY), and a static insurance comparison table.
- **Roadmap post-hackathon:** XGBoost ranking model, Gemini-powered multilingual chatbot, full state-by-state scheme database, and application tracking with document checklist automation.

9. Technology used for every part of the System

Layer	Technology	Why It Was Chosen
Frontend	React, Tailwind CSS	Fast, component-driven UI with rapid, consistent styling
Backend	Fast API	Lightweight python backend exposing REST endpoints for profile intake , eligibility scoring and insurance comparison
Database	Supabase Postgre SQL/ SQLite	Structured , relational storage for scheme rules , user profile and insurance data
Authentication	Firebase Authentica tion	Secure, trusted user login and session management
Artificial Intelligence	Gemini API	Powers recommendations, document simplification, and chat
OCR	Tesseract OCR	Accurate, open-source text extraction from scans and PDFs
Deployment	Google Cloud	Scalable, independent hosting for frontend and backend

10. Alignment with Sustainability Goals

Sustainability & SDG Alignment

BenefitBuddy AI is a pure information intervention that improves the efficiency of existing systems without requiring new physical infrastructure

- SDG 1 (No Poverty): Surfaces cash-transfer and subsidy schemes that households currently miss
- SDG 3 (Good Health): Prevents medical-bill bankruptcy by mapping public programs complemented by tailored micro-insurance
- SDG 10 (Reduced Inequalities): Democratizes elite-level financial consulting for rural and marginalized groups
- SDG 5. Gender Equality we help women get welfare schemes, maternity benefits and pension schemes.
- SDG 8. Decent Work & Economic Growth we help businesses and farmers get credit schemes and subsidies.
- SDG 9. Industry, Innovation & Infrastructure we build on India's public infrastructure instead of making a new one.

11. Conclusion

Benefit Buddy AI closes the gap between the welfare and insurance a citizen is entitled to and what they actually receive — not by building new infrastructure, but by making one profile do the work of a dozen scattered portals. The prototype demonstrates the core loop: profile intake, transparent eligibility matching with plain-language reasoning, and a live insurance comparator spanning government and private plans. What remains — multilingual voice support, a broader scheme database, and application tracking — is a natural extension of an architecture that's already built to support it. For the millions who miss out simply because they didn't know, BenefitBuddy AI turns "you may qualify"

into "here's exactly what you get, and why."